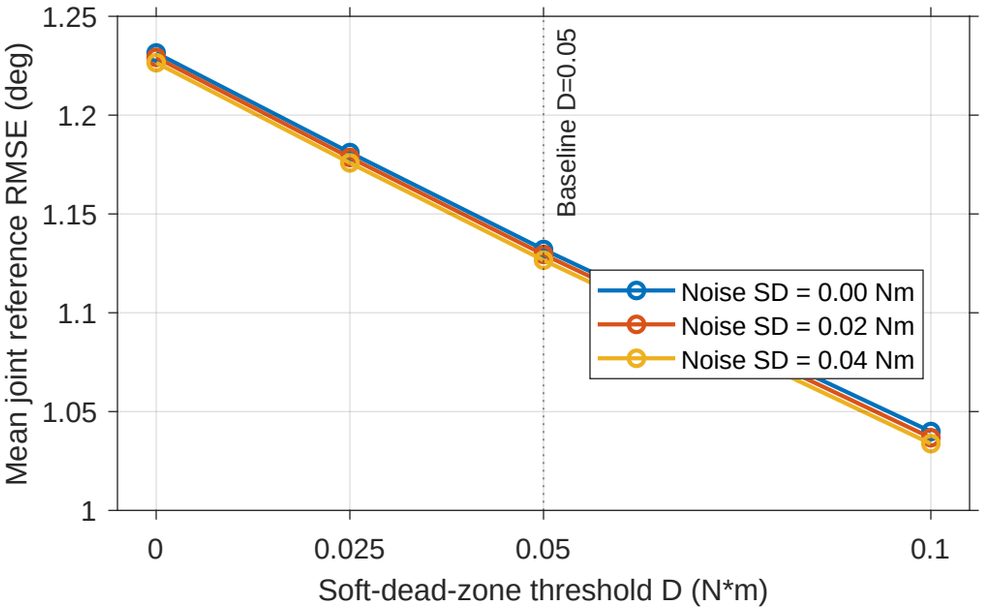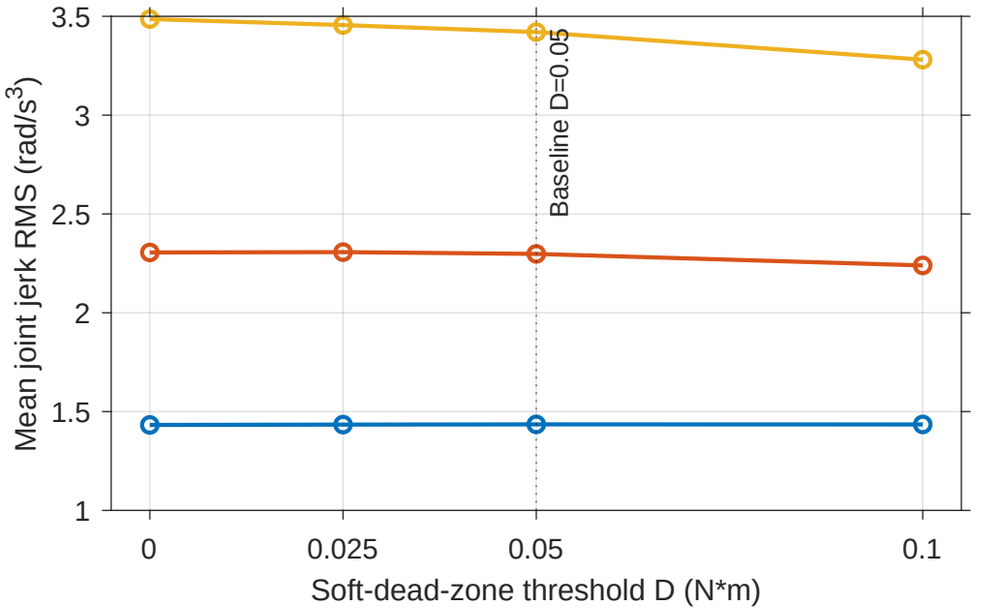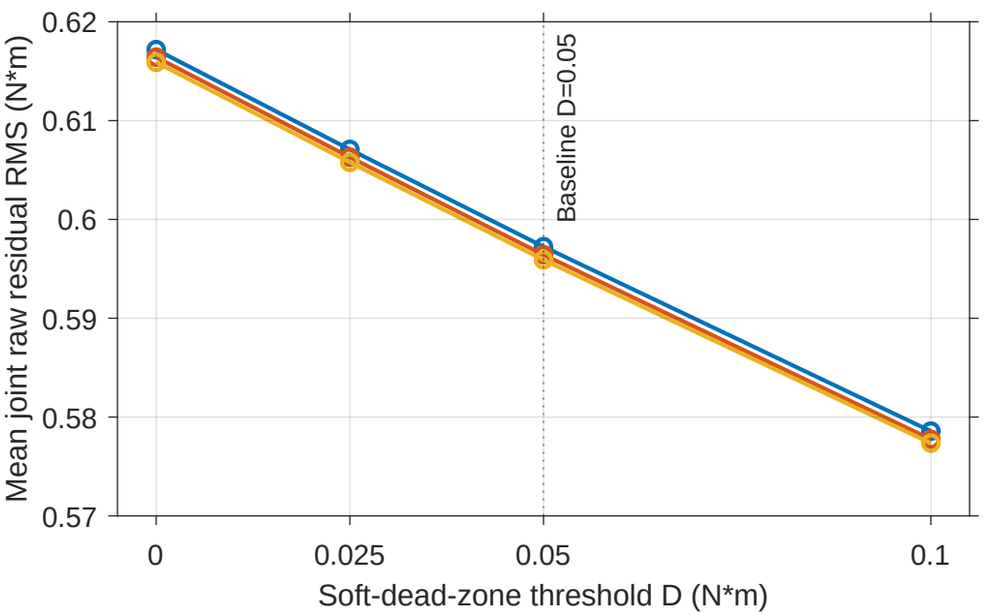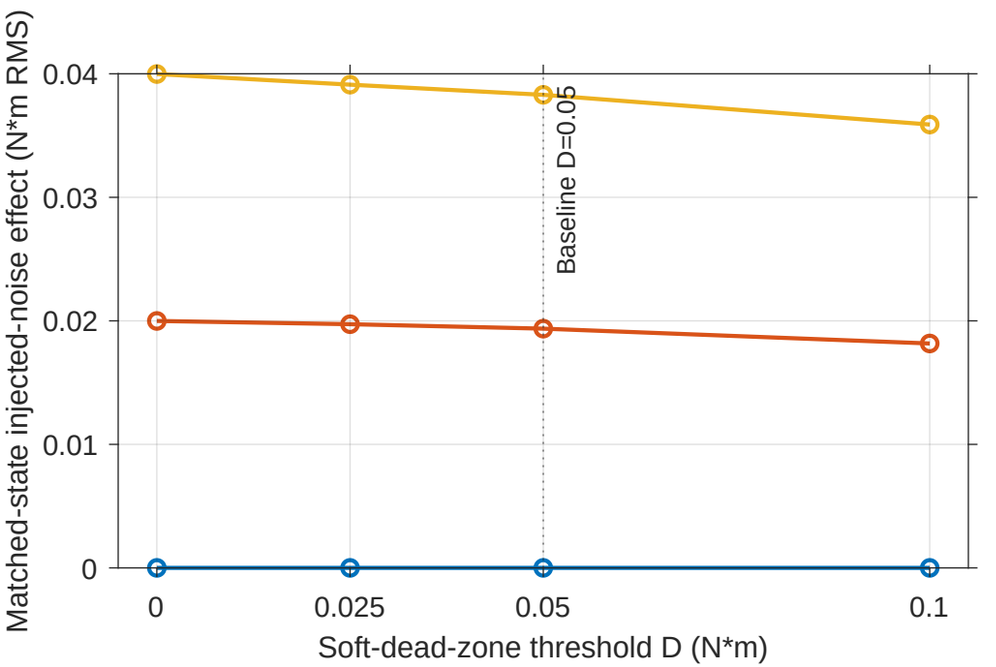Dead-zone threshold and simulated feedback-noise sensitivity